Practice Set 2 Paper 1 – Basic Sciences in Urology

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Practice paper compiled from the question bank.

Q1. Describe anatomy of pelvic floor in females. Discuss fertility sparing treatment options for procedentia.

Type: LAQ (25 marks) (pelvic floor anatomy also answered in 2017-Winter Paper 3, Q2 and 2015-Summer Paper 1, Q2)

"Procedentia" is read as procidentia (complete uterine prolapse); "fertility sparing" means uterus-preserving treatment for a woman who wants future pregnancy.

A. Anatomy of the female pelvic floor

Support comes from the **levator ani**, the **endopelvic fascia and ligaments**, and the **perineal membrane and perineal body**, anchored to the bony pelvis.

1. Pelvic diaphragm

Muscle	Attachment	Function
Puborectalis	Back of pubis, loops behind anorectal junction	Anorectal angle
Pubococcygeus	Pubis and arcus tendineus to viscera and anococcygeal raphe	Slings urethra and vagina
Iliococcygeus	Arcus tendineus and ischial spine to coccyx	Horizontal **levator plate**
Coccygeus (not levator)	Ischial spine to sacrum and coccyx	Lies on sacrospinous ligament

- Nerve to levator ani **(S3–S5)**; **pudendal nerve (S2–S4)** to sphincters and perineal muscles.
- Tonic contraction closes the **urogenital hiatus**; the uterus and upper vagina rest on the levator plate.

2. Perineal membrane and perineal body

- The membrane spans the pubic arch, pierced by urethra and vagina; above it lies the striated urogenital sphincter. The perineal body anchors the distal vagina.

3. Ligaments and surgical anchors

- **Cardinal ligament** (side wall to cervix; carries the uterine artery) and **uterosacral ligament** (sacrum to cervix). The ureter is 4 cm from the uterosacral ligament near the sacrum but 0.9 cm near the cervix.
- Pubocervical fascia attaches to the **ATFP (arcus tendineus fasciae pelvis)**, pubis to ischial spine.
- **Sacrospinous ligament (SSL)**, 7–8 cm, ischial spine to sacrum: the pudendal nerve and vessels pass behind the spine, the sacral plexus just posterior.
- **Sacral promontory**: left common iliac vein and right ureter each about 2.7 cm away.

4. DeLancey levels (1992)

Level	Structures	Loss produces
I suspension	Cardinal–uterosacral complex	**Uterine or vault prolapse**, enterocele
II attachment	Paravaginal attachment to ATFP	Anterior or posterior wall prolapse
III fusion	Perineal membrane, perineal body	Urethral hypermobility, distal rectocele

5. Why prolapse occurs

- Muscle fails first (birth trauma, levator avulsion, denervation). The hiatus widens, the ligaments carry the whole load and stretch. Collagen disorders (Ehlers–Danlos, Marfan) add risk; also vaginal birth, forceps, age, obesity, constipation.

Diagram to draw: sagittal female pelvis with levator plate and DeLancey levels I–III.

B. Procidentia: definition and assessment

- **Procidentia**: the uterus lies wholly outside the hymen, **POP-Q (pelvic organ prolapse quantification) stage 4**.
- Assess all compartments, occult SUI (stress urinary incontinence) with prolapse reduced, PVR (post-void residual), hydronephrosis, cervical length and ulceration. Abnormal bleeding needs gynaecological evaluation first.

C. Fertility-sparing options

Principle: restore **level I** support while keeping uterus and cervix. Defer definitive surgery until the family is complete where possible, because pregnancy and delivery cause recurrence.

1. Conservative (first line)

- PFMT (pelvic floor muscle training), weight loss, treat constipation.
- **Pessary**: space-filling (Gellhorn, cube) for procidentia; removed for conception; review about 6-monthly for discharge and erosion.

2. Uterus-preserving surgery

Operation	Technique	Fertility points
Sacrospinous hysteropexy (vaginal)	Cervix sutured to the SSL **1.5–3 cm medial to the ischial spine**, into the ligament	Non-inferior to vaginal hysterectomy with uterosacral suspension (SAVE U trial); buttock pain, pudendal injury
Uterosacral hysteropexy (vaginal or laparoscopic)	Shorten and reattach uterosacral ligaments to cervix, sutures lateral to medial	No mesh; cystoscopy for ureteric kinking
Sacrohysteropexy (open, laparoscopic, robotic)	Mesh from posterior cervix to anterior longitudinal ligament over the promontory	Most durable apical support; later pregnancy usually by caesarean
Manchester (Fothergill)	Cervical amputation, cardinal ligaments sutured in front of the stump, anterior colporrhaphy	Suits elongated cervix, but risks **cervical stenosis, incompetence, miscarriage, preterm labour**: avoid if pregnancy wanted

- **Also know:** Indian abdominal slings for nulliparous prolapse (Shirodkar, Purandare, Khanna, Virkud). **Le Fort colpocleisis** keeps the uterus but not fertility (elderly only).

3. Choosing

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Procidentia, wants pregnancy -> PFMT + pessary
  -> fails -> sacrospinous hysteropexy or
     laparoscopic sacrohysteropexy (avoid Manchester)
  -> recurrence after childbearing -> hysterectomy
     + vault suspension or sacrocolpopexy
```

D. Complications and follow-up

- Recurrence (mostly posterior), de novo SUI, mesh exposure (3.4–5.4% after sacrocolpopexy), ureteric kinking. The retained uterus needs cervical screening.

> **Key points to remember:**
> - Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
> - DeLancey I (uterosacral–cardinal) failure causes uterine prolapse.
> - Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
> - Manchester amputates the cervix: poor choice for future pregnancy.

Q2. Describe embryology & clinical implications of IVC anomalous development.

Type: LAQ (25 marks) (retrocaval ureter also answered in 2017-Winter Paper 3, Q3d)

Introduction

- The **IVC (inferior vena cava)** forms between the 6th and 8th weeks from three pairs of fetal veins by anastomosis, regression and persistence. The final IVC is **right-sided**, so long left veins must cross the aorta to reach it.
- Most anomalies are silent CT (computed tomography) findings. They **bleed at surgery, mimic lymph nodes, trap the ureter and alter IVC filter placement**.

Embryology

Three paired venous systems

Vein	Position	Adult derivatives
Posterior cardinal (first)	Dorsolateral	Iliac confluence; mostly regresses
Subcardinal	Ventromedial	Right: **suprarenal IVC**; intersubcardinal anastomosis: **left renal vein**; left: adrenal and gonadal veins
Supracardinal (last)	Dorsomedial	Right: **infrarenal IVC**; thoracic parts: **azygos and hemiazygos**

Segments of the definitive IVC (top to bottom)

1. **Hepatic**: right vitelline vein.
2. **Suprarenal**: right subcardinal vein (via the subcardinal–hepatic anastomosis).
3. **Renal**: subcardinal–supracardinal anastomosis.
4. **Infrarenal**: right supracardinal vein.
5. **Iliac confluence**: posterior cardinal veins.

Renal collar

- A venous ring encircles the aorta at renal level. The **ventral (pre-aortic) limb** persists as the left renal vein; the **dorsal (retro-aortic) limb** regresses.
- The right supracardinal vein lies **dorsal** to the ureter, so the normal ureter lies in front of the IVC.

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Left-sided channels regress -> right-sided persist
  -> single right IVC + pre-aortic left renal vein
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Diagram to draw: three stages: paired cardinal veins, anastomoses with renal collar, definitive IVC.

Anomalies

Anomaly	Embryological error	Key features
Retrocaval ureter ("preureteral vena cava")	Infrarenal IVC forms from a vein **ventral** to the ureter	Right only; 1 in 1500 autopsies; male 3:1; presents 3rd–4th decade. **Type I** low loop, fishhook, obstructs; **type II** high loop, sickle, mild
Circumaortic left renal vein	Dorsal limb persists with ventral	Commonest (2–16%); posterior limb runs lower into lumbar IVC
Retroaortic left renal vein	Ventral limb regresses, dorsal persists	Less common; may be compressed ("posterior nutcracker")

Anomaly	Embryological error	Key features
Left IVC	Left supracardinal persists, right regresses	Crosses to the right at renal level, in front of the aorta
Duplicated IVC	Both supracardinals persist	Left IVC joins left renal vein; double right cava can trap the ureter
Azygos continuation	Subcardinal–hepatic anastomosis fails	Large azygos; hepatic veins drain directly to right atrium; polysplenia, heterotaxy
Absent infrarenal IVC	Supracardinal and posterior cardinal failure	Deep vein thrombosis in the young

- For retrocaval ureter Campbell names a persistent **lumbar subcardinal** (Chapter 52) or **posterior cardinal** vein (Chapter 93, most texts); both mean a ventral vein forms the IVC.

Clinical implications

1. Diagnosis

- Venous-phase CT or MRI (magnetic resonance imaging). A left or duplicated IVC mimics **para-aortic lymphadenopathy**; a large azygos mimics a mediastinal mass, risking needless biopsy.
- Retrocaval ureter: S-shaped or fishhook ureter at L3–L4 on retrograde pyelography; CT urography confirms; diuretic renography grades obstruction.

2. RPLND (retroperitoneal lymph node dissection)

- Preoperative CT must look for circumaortic or retroaortic veins and retrocaval ureter. A left or duplicated IVC lies in the left template; a retroaortic limb tears when lumbar vessels are divided.

3. Radical nephrectomy

- The hidden posterior limb of a circumaortic vein bleeds when the artery is approached from behind: ligate both limbs. Caval thrombus in a duplicated IVC needs rerouting of the left IVC.

4. Living donor nephrectomy

- The left kidney is preferred for its long vein. A circumaortic or retroaortic vein shortens it and adds bleeding risk; CT angiography is mandatory, and the right kidney may be chosen.

5. Retrocaval ureter repair

- Observe non-obstructed type II. For obstruction, **divide the ureter, transpose it in front of the IVC and re-anastomose** over a stent, excising the aperistaltic segment; laparoscopic or robotic. A pyeloplasty that misses it fails.

6. Interventional and vascular

- **IVC filter:** with duplication, emboli bypass a single filter, so filter both cavae or place it suprarenally.
- Aortic surgery risks retroaortic vein injury. A retroaortic vein can cause haematuria or left varicocele.
- **Also know:** variable left adrenal vein entry (adrenal vein sampling); right gonadal vein into right renal vein in up to 10%.

> **Key points to remember:**
>
> - Infrarenal IVC = right supracardinal; suprarenal = right subcardinal; hepatic = vitelline.
> - Circumaortic left renal vein is commonest (2–16%): ligate both limbs.
> - Retrocaval ureter: right only, 1 in 1500; transpose anterior to IVC.
> - Duplicated IVC: filter both cavae; do not mistake it for nodes.

Q3a. Patient positioning in lap urology

Type: SAQ (10 marks) (pneumoperitoneum physiology also answered in 2019-Summer Paper 1, Q2)

Principles

- Positioning uses **gravity** to move bowel away and gives port or robot access while keeping ventilation safe.
- Injury comes from **stretch, compression or ischaemia**, so position and duration act together; nerve injury occurs in about 2.8%. Shared team responsibility (ASA [American Society of Anesthesiologists] advisory, 2000).
- Secure IV (intravenous) access, oximetry and BP (blood pressure) cuff before final positioning: tucked arms become inaccessible.

Positions

Position	Used for	Set-up	Hazards
Steep Trendelenburg, low lithotomy or split legs	RARP (robot-assisted radical prostatectomy), cystectomy	Arms tucked and padded; chest strap over egg-crate; **no shoulder braces**; tilt about 22–30°	Brachial plexus; femoral nerve (excess hip flexion); facial and laryngeal oedema; corneal abrasion; tube into right bronchus
Modified flank (30–45°)	Transperitoneal nephrectomy, pyeloplasty	Back on gel rolls; upper arm across chest; wide tape; tilt to test	Pressure points, slippage
Full lateral (90°), table flexed	Retroperitoneoscopy, adrenalectomy	**Axillary roll** 3–4 fingerbreadths below axilla; lower leg flexed 45°, upper straight on pillows	**Rhabdomyolysis**; sciatic stretch
Prone	Posterior adrenalectomy	Pad torso, elbows, hips	Airway, face and eyes

Rhabdomyolysis

- About 0.4% (1% retroperitoneal). Risk: men, BMI (body mass index) ≥25, **kidney rest**, full flexion, long cases. Downside hip pain, brown urine, CK (creatine kinase) >5,000 units. Prevent: limit kidney rest (Campbell gives 20–30 minutes in one place, under 1 hour in another), avoid hypotension, keep cases under 5 hours.

Prevention and physiology

- Abduction ≤90°; pad ulnar groove and fibular head; recheck after every table move.
- **Head-down** raises cardiac output but splints the diaphragm; restrict fluids.
- Nerve injury on day 1: neurology review, conduction studies, physiotherapy.
- **Also know:** forearm compartment syndrome from infiltrated IV lines; Xi side-docking avoids stirrups.

Diagram to draw: full flank position with axillary roll and table break.

Key points to remember:

- Nerve injury about 2.8%; brachial plexus is commonest at risk.
- Never shoulder braces in steep Trendelenburg.
- Axillary roll below, not in, the axilla.
- Rhabdomyolysis: men, BMI ≥25, kidney rest; CK >5,000.

Q3b. Familial malignancies in Urology.

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 2, Q3f)

Principles

- **Knudson two-hit hypothesis:** every cell carries one inherited mutant tumour-suppressor allele, so one more hit causes cancer: tumours are **early, bilateral, multifocal**.
- **4–6% of RCC (renal cell carcinoma)** is familial. All familial RCC syndromes are **autosomal dominant**; only HPRC (hereditary papillary renal carcinoma) involves an activated oncogene.

Hereditary kidney cancer syndromes

Syndrome	Gene (chromosome)	Kidney tumour	Other features
VHL (von Hippel–Lindau)	VHL (3p25)	Clear cell, multifocal	Retinal angioma, CNS (central nervous system) haemangioblastoma, phaeochromocytoma
HPRC	MET (7q31)	Multiple bilateral papillary	None
HLRCC (hereditary leiomyomatosis and RCC)	FH (fumarate hydratase, 1q42–43)	**Aggressive** papillary, collecting duct	Skin and uterine leiomyomas
BHD (Birt–Hogg–Dubé)	FLCN (17p11.2)	Chromophobe, hybrid oncocytic	Fibrofolliculomas, lung cysts, pneumothorax
TSC (tuberous sclerosis complex)	TSC1 (9q34), TSC2 (16p13.3)	Multiple AMLs (angiomyolipomas); RCC 2–3%	Angiofibromas, cortical tubers

- **Also know:** SDH (succinate dehydrogenase)-deficient RCC; BAP1 (uveal melanoma, mesothelioma); PTEN (Cowden).

Kidney management

- **3 cm rule:** watch tumours <3 cm; nephron-sparing surgery at ≥3 cm (VHL, HPRC, BHD) to save function for repeat surgery. **Exception: HLRCC**: excise promptly, wide margins.
- TSC AML: everolimus at 3–5 cm; embolise or excise ≥5 cm. **Belzutifan** (HIF-2α [hypoxia-inducible factor 2α] inhibitor) approved 2021 for VHL.

Other urological cancers

- **Prostate:** BRCA2, BRCA1, ATM, CHEK2, PALB2, HOXB13. Father and brother affected: 9-fold risk. Earlier screening, cautious active surveillance, **PARP (poly ADP-ribose polymerase) inhibitors**.
- **UTUC (upper tract urothelial carcinoma): Lynch syndrome** (MLH1, MSH2, MSH6, PMS2): about 14-fold risk, 75-fold with MSH2. MMR (mismatch repair) immunohistochemistry, then germline testing.
- **Testis:** affected brother RR (relative risk) 8–12, father 2–4.
- **Phaeochromocytoma:** about one-third hereditary: MEN-2 (multiple endocrine neoplasia type 2; RET), VHL, NF1 (neurofibromatosis type 1), SDHB (highest malignancy).
- **Also know:** Wilms tumour (WT1: WAGR [Wilms, aniridia, genitourinary anomalies, retardation] and Denys–Drash; Beckwith–Wiedemann).

Approach

- Young age, multifocality or family history: genetic counselling, germline testing, screen relatives.

Diagram to draw: VHL loss → HIF accumulation → VEGF (vascular endothelial growth factor) → angiogenesis.

> **Key points to remember:**
>
> - All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
> - VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
> - Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

Q3c. Lymphogranuloma venereum

Type: SAQ (10 marks)

Definition

- **LGV (lymphogranuloma venereum)** is an invasive, lymphotropic STD (sexually transmitted disease) caused by ***Chlamydia trachomatis*** **serovars L1, L2 and L3**.
- Of 15 serovars, A–C cause trachoma and D–K anogenital infection. The L serovars invade lymphatics, causing lymphadenitis and scarring.

Epidemiology

- Endemic in the tropics, including India; increasing as **proctitis in MSM (men who have sex with men)**. Not more frequent or virulent in HIV (human immunodeficiency virus).

Clinical stages

Stage	Timing	Features
Primary	3–30 days	Small painless papule or ulcer, often unnoticed. Rectal inoculation → **proctitis** (bleeding, pain, tenesmus, discharge), now the commonest presentation
Secondary (inguinal syndrome)	2–6 weeks later	Fever, malaise; **painful, matted, unilateral inguinal or femoral buboes** that may form fistulae. **Groove sign**: nodes above and below the inguinal ligament
Tertiary	Untreated	Fibrosis and lymphatic block: **genital elephantiasis** (esthiomene in women), perirectal abscess, fistulae, **rectal stricture**

- Inguinal disease is commoner in men: the cervix and vagina drain to pelvic nodes.

Differential diagnosis of genital ulcer

- Syphilis (painless ulcer and nodes); herpes (painful vesicles); chancroid (painful ulcer, suppurative nodes); donovanosis (beefy ulcer, no nodes); **LGV: transient painless lesion, then painful matted nodes**.

Investigations

- **NAAT (nucleic acid amplification test)** of ulcer, rectal or node aspirate; LGV-specific genotyping confirms where available. Rectal chlamydia NAAT in all proctocolitis.
- Serology not routine. Screen for HIV, syphilis, gonorrhoea; offer PrEP (pre-exposure prophylaxis).

Treatment

- **Doxycycline 100 mg twice daily for 21 days** (first line).
- Alternatives: erythromycin 500 mg four times daily for 21 days, or azithromycin 1 g weekly for 3 weeks (pregnancy; consider test of cure by NAAT).
- Fluctuant buboes: aspirate through healthy skin to prevent sinus formation.
- **Late disease:** stricture dilatation or resection, fistula repair, reduction of elephantiasis.

Diagram to draw: groin with buboes above and below the inguinal ligament (groove sign).

Key points to remember:

- Serovars L1–L3; invasive and lymphotropic.
- Proctitis is now the commonest presentation.
- Groove sign; esthiomene; rectal stricture.
- Doxycycline 100 mg twice daily for 21 days.

Q3d. Isotope studies of urology

Type: SAQ (10 marks) (renal scans in depth: see Practice Set 2 Paper 1, Q3e; diuretic renography: see Practice Set 2 Paper 1, Q3f)

Principles

- Nuclear medicine shows **function, not structure**: a tracer in tiny dose follows a physiological or molecular process.
- **Gamma camera** (sodium iodide crystal, photomultipliers, collimator): planar (about 20 mm resolution) or **SPECT (single-photon emission computed tomography)** with CT (about 10 mm).
- **PET (positron emission tomography):** positron annihilation gives two 511 keV photons 180° apart, detected in coincidence: sharper (about 5 mm), less noise, quantitative SUV (standardised uptake value).
- **^{99m}Tc (technetium-99m):** 6-hour half-life, 140 keV photon: the workhorse. ^{18}F: 110 minutes, sharpest PET image.

Applications

Area	Tracer	Use
Kidney	^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid), -MAG3 (mercaptoacetyltriglycine), -DMSA (dimercaptosuccinic acid)	GFR (glomerular filtration rate), split function, obstruction, scars, transplant
Bone	^{99m}Tc-MDP (methylene diphosphonate)	High-risk prostate cancer staging; low specificity
Prostate	**PSMA (prostate-specific membrane antigen) PET**; choline; fluciclovine	PSMA preferred (NCCN [National Comprehensive Cancer Network]) for staging and recurrence; specificity >90%
Therapy	^{177}Lu-PSMA-617; radium-223	mCRPC (metastatic castration-resistant prostate cancer); VISION trial (2021)
FDG (fluorodeoxyglucose) PET	Glucose analogue trapped in cells	**Residual seminoma >3 cm**; penile cancer with palpable nodes; distant urothelial disease. Poor in prostate and clear cell RCC (renal cell carcinoma)
Renal mass	Sestamibi SPECT; girentuximab PET	Oncocytoma "hot" on sestamibi; clear cell RCC "hot" on girentuximab
Adrenal	^{123}I-MIBG (metaiodobenzylguanidine), ^{68}Ga-DOTATATE, FDG	Metastatic or extra-adrenal phaeochromocytoma; MIBG best in MEN-2 (multiple endocrine neoplasia type 2)
Sentinel node	^{99m}Tc-nanocolloid plus blue dye	Penile cancer, impalpable nodes
Scrotum	^{99m}Tc-pertechnetate	Torsion cold, epididymitis hot
Reflux	Direct radionuclide cystography	VUR (vesicoureteral reflux) follow-up: more sensitive, lower dose than VCUG (voiding cystourethrogram)
Infection	^{111}In-labelled leucocytes; ^{67}Ga	Occult abscess; not in transplant

Diagram to draw: gamma camera chain: patient, collimator, crystal, photomultiplier, image.

Key points to remember:

- ^{99m}Tc: 6 hours, 140 keV; PET: two 511 keV photons.
- DTPA filtered, MAG3 secreted, DMSA cortical.
- PSMA PET preferred; FDG for residual seminoma >3 cm.
- MIBG best in MEN-2; sentinel node with ^{99m}Tc-nanocolloid.

Q3e. Renal isotope scans

Type: SAQ (10 marks) (also answered in 2017-Winter Paper 2, Q3f and 2021-Summer Paper 1, Q3e; diuretic protocol: see Practice Set 2 Paper 1, Q3f)

Principle

- Each tracer images the part of the nephron that handles it. Plasma clearance is about 20% filtration and 80% tubular secretion.

Tracers

Feature	^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid)	^{99m}Tc-MAG3 (mercaptoacetyltriglycine)	^{99m}Tc-DMSA (dimercaptosuccinic acid)
Handling	Glomerular filtration only; poor images in renal failure	Protein-bound; tubular secretion; good images in renal failure	Filtered, then retained by proximal tubules
Study	Dynamic	Dynamic	**Static cortical**, 2 hours
Main use	**GFR (glomerular filtration rate)**; captopril study	**Split function, obstruction (first choice)**; transplant	**Pyelonephritis and scars**
Dose	5–10 mCi	5–10 mCi	5 mCi adult; 50 µCi/kg child

Dynamic renography: method

- Hydrate; catheter for outlet obstruction; cap nephrostomy; IV (intravenous) bolus.
- **Perfusion phase:** frames every 1–2 seconds; kidney appears within seconds of the aorta.
- **Function phase:** regions of interest with background subtraction. **Split function = area under each curve at 1–3 minutes**; normal up to 45:55.
- **Drainage phase:** MAG3 peaks at 3–5 minutes, bladder by 10–15 minutes; a rising curve needs furosemide (see Q3f).
- **GFR:** DTPA camera method (Gates) or plasma-sample clearance.

DMSA cortical scan

	Acute pyelonephritis	Scar
Defect	Ill-defined	Sharp, cortical thinning
Kidney	Normal size	Often small
At 6 months	Resolves	Persists

Special studies

- **Captopril renography (DTPA):** the stenotic kidney keeps GFR by angiotensin II efferent constriction; captopril removes it, so uptake falls. Repeat off ACE (angiotensin-converting enzyme) inhibitors: improvement **≥10%** means renovascular hypertension likely. Largely replaced by CT (computed tomography) angiography.
- **Transplant (MAG3):** acute rejection: perfusion falls and worsens; delayed graft function: perfusion kept, improves. Shows urine leak.

Advantages and limits

- Quantitative, low radiation, safe in contrast allergy and pregnancy (MAG3 fetal dose <5 mGy); poor anatomy.

Diagram to draw: renogram: perfusion spike, peak at 3–5 minutes, washout, 1–3 minute split window.

Key points to remember:

- DTPA = GFR; MAG3 = obstruction and transplant; DMSA = scars.
- Split function from 1–3 minute counts; normal 45–55%.
- Captopril renogram: ≥10% improvement off ACE inhibitors.
- Scar persists at 6 months; pyelonephritis resolves.

Q3f. Diureteric renography

Type: SAQ (10 marks) (PUJ obstruction renography also in 2021-Summer Paper 1, Q3e; tracers: see Practice Set 2 Paper 1, Q3e)

"Diureteric" is read as diuretic renography.

Principle

- **Dilatation is not obstruction.** A capacious pelvis retains tracer passively and furosemide washes it out; an obstructed system cannot empty even under diuresis.
- **MAG3 (technetium-99m mercaptoacetyltriglycine)** is the tracer of choice (good curves even with poor function); DTPA (diethylenetriaminepentaacetic acid) is an alternative.

"Well-tempered" protocol

Element	How	Why
Hydration	Oral; IV (intravenous) fluid in children (about 15 mL/kg)	Dehydration mimics obstruction
Bladder catheter	Urethral catheter	Removes full-bladder back-pressure and VUR (vesicoureteral reflux) effects
Furosemide	0.5 mg/kg, adult maximum 40 mg; 1 mg/kg in children	Maximal diuresis
Timing	**F+20** (classic, pelvis full); F+15 (Campbell authors); **F0** (with tracer); **F−15** (already in diuresis)	Early diuretic into an unfilled system misleads; always state timing
Regions of interest	Cortex for split; dilated pelvis for drainage	Wrong placement gives spurious values

- First study after about 4–6 weeks of age (immature function).

Outputs

- **DRF (differential renal function)** from 1–3 minute counts: normal 45–55%.
- **T½ (half-clearance time)**, from maximal filling, not from injection:

T½ after furosemide	Interpretation
<10 minutes	Not obstructed
10–20 minutes	Indeterminate
>20 minutes	Obstructed

- **O'Reilly curves:** I normal; II obstructed (rising); IIIa dilated, not obstructed (falls after furosemide); IIIb equivocal. A **biphasic curve** and poor upright (post-void) drainage predict progression.

Indications

- PUJ (pelviureteric junction) obstruction, antenatal hydronephrosis, megaureter, after pyeloplasty, transplant ureter.
- Consider surgery for **DRF <40%**, falling DRF, T½ >20 minutes or upright retention, with symptoms.

Pitfalls

- Dehydration, full bladder, poor function, very capacious pelvis, wrong ROI (region of interest), mistimed diuretic.

- No agreed paediatric T½ cut-off; neonatal false-positive rates reach 44%, so **serial trends and curve shape** decide. Equivocal: Whitaker test or MRU (magnetic resonance urography).

Diagram to draw: O'Reilly curves I, II, IIIa and IIIb with furosemide arrow.

Key points to remember:

- Well-tempered: hydrate, catheter, standard diuretic dose and timing.
- Furosemide 0.5 mg/kg (adult maximum 40 mg); 1 mg/kg in children.
- T½ <10 not obstructed; 10–20 indeterminate; >20 obstructed.
- Surgery: DRF <40%, falling DRF, or symptoms.

Rapid Revision Sheet – Practice Set 2 Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe anatomy of pelvic floor in females. Discuss fertility sparing treatment options for procedentia. (LAQ)

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
- DeLancey I (uterosacral–cardinal) failure causes uterine prolapse.
- Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
- Manchester amputates the cervix: poor choice for future pregnancy.

Q2. Describe embryology & clinical implications of IVC anomalous development. (LAQ)

- Infrarenal IVC = right supracardinal; suprarenal = right subcardinal; hepatic = vitelline.
- Circumaortic left renal vein is commonest (2–16%): ligate both limbs.
- Retrocaval ureter: right only, 1 in 1500; transpose anterior to IVC.
- Duplicated IVC: filter both cavae; do not mistake it for nodes.

Q3a. Patient positioning in lap urology (SAQ)

- Nerve injury about 2.8%; brachial plexus is commonest at risk.
- Never shoulder braces in steep Trendelenburg.
- Axillary roll below, not in, the axilla.
- Rhabdomyolysis: men, BMI ≥25, kidney rest; CK >5,000.

Q3b. Familial malignancies in Urology. (SAQ)

- All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
- VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
- Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

Q3c. Lymphogranuloma venereum (SAQ)

- Serovars L1–L3; invasive and lymphotropic.
- Proctitis is now the commonest presentation.
- Groove sign; esthiomene; rectal stricture.
- Doxycycline 100 mg twice daily for 21 days.

Q3d. Isotope studies of urology (SAQ)

- ^{99m}Tc: 6 hours, 140 keV; PET: two 511 keV photons.
- DTPA filtered, MAG3 secreted, DMSA cortical.
- PSMA PET preferred; FDG for residual seminoma >3 cm.
- MIBG best in MEN-2; sentinel node with ^{99m}Tc-nanocolloid.

Q3e. Renal isotope scans (SAQ)

- DTPA = GFR; MAG3 = obstruction and transplant; DMSA = scars.
- Split function from 1–3 minute counts; normal 45–55%.
- Captopril renogram: ≥10% improvement off ACE inhibitors.
- Scar persists at 6 months; pyelonephritis resolves.

Q3f. Diureteric renography (SAQ)

- Well-tempered: hydrate, catheter, standard diuretic dose and timing.
- Furosemide 0.5 mg/kg (adult maximum 40 mg); 1 mg/kg in children.
- T½ <10 not obstructed; 10–20 indeterminate; >20 obstructed.
- Surgery: DRF <40%, falling DRF, or symptoms.